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Material-specific interference control is dissociable and lateralized in human prefrontal cortex

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ABSTRACT

The prefrontal cortex (PFC) plays a key role in the ability to pursue a particular goal in the face of competing alternatives, an ability that is fundamental to higher-order human behavior. Whether this region contributes to cognitive control through material-general mechanisms, or through hemispheric specialization of component abilities, remains unclear. Here we show that left or right ventrolateral PFC damage in humans leads to doubly dissociable deficits in two classic tests of interference control. Patients with damage centered on left ventrolateral prefrontal cortex had exaggerated interference effects in the color-word Stroop, but not the Eriksen flanker task, whereas patients with damage affecting right ventrolateral prefrontal cortex showed the opposite pattern. Thus, effective interference resolution requires either right or left lateral PFC, depending on the nature of the task. This finding supports a lateralized, material-specific account of cognitive control in humans.

1. Introduction

1.1. Theoretical underpinnings and clinical relevance of interference control

Resistance to interference, defined as "the ability to ignore or inhibit irrelevant information while executing a plan", is essential for successful human behavior (Dempster and Corkill, 1999, p. 397). Research on interference control spans over a hundred years and permeates the field of psychology: changes in susceptibility to interference have been used to explain development of cognitive abilities, age-related cognitive decline and psychiatric disorders including attention-deficit/hyperactivity disorder, obsessive-compulsive disorder, autism, and schizophrenia (Barkley, 1997; Ciesielski and Harris, 1997; Darowski et al., 2008; Dempster, 1995; Enright and Beech, 1993; Nestor and O'Donnel, 1998; Ridderinkhof et al., 1997). A central question in the study of interference control and, more generally, executive function remains: is resistance to interference a unitary, domain-general function or is it instead supported by dissociable, material-specific processes? This fundamental question has important implications for the way we conceptualize, diagnose, assess, and treat disorders of cognitive control.

1.2. Support for interference control as a domain-general process

Behavioral studies in healthy subjects suggest a close relationship between different forms of resistance to interference. A classic paper by Friedman and Miyake (2004) sought to address the unity and diversity among tasks of interference control, including two of the most commonly used tests in clinical psychology and cognitive neuroscience, the Stroop (Stroop, 1935) and Eriksen flanker tasks (Eriksen and Eriksen, 1974). Using latent variable analysis of individual differences, they showed a tight relationship between tasks requiring suppression of prepotent verbal responses, whether triggered by verbal (Stroop interference) or spatial (flanker interference) distractors. This raises the possibility of a common neural mechanism for these two forms of interference control. Functional imaging and single-unit studies argue that a suite of frontal and parietal brain regions alternatively termed multiple-demand (Duncan, 2001), task-positive (Fox et al., 2005), or the cognitive control network (Cole et al., 2013; Cole and Schneider, 2007) is broadly engaged across a range of tasks that require cognitive flexibility. This frontoparietal network includes

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ventrolateral prefrontal cortex (VLPFC) and is engaged by tasks requiring interference control, including Stroop (Duncan, 2010; Fedorenko et al., 2013). A different line of inquiry also supports the functional generality of prefrontal cortex (PFC); single-unit recordings in primate VLPFC show that these neurons can flexibly code task-relevant information (Asaad et al., 2000; Cromer et al., 2010; Freedman et al., 2001; Fuster et al., 2000). In sum, this work argues that interference control is domain-general, with VLPFC part of a network that is engaged to implement control across diverse tasks.

1.3. Evidence for multiple-process and material-specific accounts of interference control

Others have argued that interference control involves multiple processes, relying on distinct PFC regions (Petrides, 2005; Stuss et al., 1995). This is supported by human functional imaging and lesion studies that show dissociable PFC contributions to interference control. Work by Kok (1999) points to multiple cortical and subcortical systems that interact over different time-scales from the processing of early sensory information to response-selection. Lesion studies also support the existence of dissociable component processes in interference control: damage to the right frontal operculum results in dissociable deficits in attentional capture and stimulus–response conflict (Michael et al., 2014). This finding shows the independence of sensory and motor processing stages in interference control. Congruent with this hypothesis, a quantitative meta-analysis of 47 fMRI studies of tasks requiring interference resolution, including Stroop and flanker tasks, showed low correlation of brain activation across tasks (Nee et al., 2007) suggesting that different forms of interference control depend on different PFC sub-regions. The authors also found hemispheric differences across tasks, with right and left lateralized foci in flanker and Stroop meta-analyses respectively; they concluded that differential patterns of activation in dorsolateral PFC (DLPFC), VLPFC, and anterior cingulate cortex (ACC) across tasks might reflect distinct cognitive processes. Taken together these findings argue against a unified network for cognitive control and instead support the existence of multiple component processes underlying resistance to interference.

Converging evidence demonstrates that flanker and Stroop tasks, despite sharing a requirement for interference control, rely on at least partly distinct, and potentially lateralized, material-specific neural substrates. Behavioral evidence shows low within subject correlation of interference control across Stroop and flanker tasks, and no interaction between these two forms of interference in a combined Stroop and flanker paradigm (Fan et al., 2003). FMRI studies show that right VLPFC (RVLPFC) activation is associated with resolution of flanker interference (Bunge et al., 2002; Hazeltine et al., 2003; Hazeltine et al., 2000; Ullsperger and von Cramon, 2001) whereas left VLPFC (LVLPFC) activation is associated with Stroop interference (Brass and von Cramon, 2004; Derrfuss et al., 2005; Leung et al., 2000; Roberts and Hall, 2008; Zysset et al., 2001). An fMRI study by Morimoto et al. (2008) observed hemispheric specialization in VLPFC for flanker interference using color words or color patches as stimuli: the color word flanker produced left VLPFC activation and the non-verbal color patch flanker activated right VLPFC (Morimoto et al., 2008). Taken together, this work fails to support the strongest forms of unitary models of interference control, instead arguing for a more functionally specific view of at least the VLPFC contribution. In contrast to fMRI work, human lesion studies can test whether a region of interest is necessary for a particular function, and so can provide stronger evidence for dissociability claims. One prior case report showed a single dissociation with interference control disrupted in a verbal, but not nonverbal task after left VLPFC damage, arguing that resistance to interference is not a common, general process engaged across tasks (Hamilton and Martin, 2005).

1.4. Aims of the present study

Here we sought evidence to disentangle competing hypotheses concerning the structure–function relationship of VLPFC and interference control. We tested whether right or left VLPFC is differentially required for the performance of two classic tests of interference control (color-word Stroop and Eriksen flanker) in patients with focal damage to these regions. If left and right VLPFC play a domain-general role in cognitive control, we would expect that damage to either region would produce either increased interference effects regardless of the task, or no effect if the intact hemisphere can fully compensate for damage in the other. On the other hand, if lateralized prefrontal regions make unique contributions to cognitive control depending on the nature of the material being processed, then the lesion–deficit relationship should be dissociable across tasks.

2. Materials and methods

2.1. Participants in Experiment 1

Eight right-handed patients with damage centered on the left ($N=3$) or right ($N=5$) VLPFC and 21 right-handed healthy controls were recruited from the McGill Cognitive Neuroscience Research Registry for the main experiment. The Edinburgh Handedness Inventory (Oldfield, 1971) was available for a subset of the total sample ($N=17$); the mean laterality inventory was 93.4 (range 80–100). The control sample included 6 men and 15 women, the LVLPFC group comprised 1 man and 2 women, and the RVLPFC sample included 1 man and 4 women. Patients were included if they had focal brain lesions affecting right or left VLPFC, as determined by structural CT and/or MRI scans. Subjects were tested at least 1 year (mean 3.3 years, range 1.2-6.4 years) after brain injury. The mean lesion volume did not differ significantly between groups (RVLPFC=67 cc, range 27–113 vs. LVLPFC 40 cc, range 13–75, $U=4.0$, $P=0.30$). Larger lesion volume did not predict greater Stroop or flanker interference effects. Individual patient lesion characteristics are described in Table 1 and Fig. 1 (A and B). One RVLPFC participant had been on a low dose of a selective serotonin reuptake inhibitor for 17 years and was euthymic. Exclusion of this subject did not alter the statistical significance of our main findings. All other patients did not take psychoactive medication and were free from psychiatric or other neurologic illness.

A summary of patient demographic information and performance on a neuropsychological battery screening for aphasia, inattention and neglect is provided in Table 2. These variables were compared across all three groups (LVLPFC, RVLPFC, Controls) using nonparametric Kruskal–Wallis H tests. Mann–Whitney U tests were applied for pairwise comparisons between patient groups. There were no significant differences between patient groups and controls with respect to age ($P=0.79$) or years of education ($P=0.96$). There were also no significant differences in forward digit span, BDI, letter cancelation tasks, sentence comprehension, or naming performance between LVLPFC and RVLPFC groups (all $P>0.10$). The LVLPFC group showed a trend towards reduced phonological fluency (FAS: $U=2.0$; $P=0.099$), semantic fluency (animals: $U=2.0$; $P=0.099$), and significantly lower backwards

Table 1
Lesion characteristics of patients in the main experiment.

Subject	*Etiology*	*Years since injury*	*Lesion volume (cc)*	*Brodmann's areas*
RVLPFC 1	Stroke	1.4	47.1	45, 44, 47, 38
RVLPFC 2	Low-grade glioma	1.3	113.1	11, 47, 20, 38, 21
RVLPFC 3	Stroke	4.4	27.2	47, 45, 46
RVLPFC 4	Stroke	4.0	78.6	45, 46, 44, 6, 9
RVLPFC 5	Meningioma resection	6.4	59.5	47, 45, 11, 46
LVLPFC 1	Stroke	3.0	30.5	44, 45, 47, 6
LVLPFC 2	Stroke	4.4	75.2	45, 47, 38, 46, 21
LVLPFC 3	Stroke	1.2	13.1	47, 45

Brodmann's areas are ordered from largest to smallest affected regions. The reported Brodmann's areas represent over 80% of the cortical lesion volume in each subject. Lesion volume was not significantly different between right and left VLPFC ($P=0.3$). RVLPFC=right ventrolateral prefrontal cortex; LVLPFC=left ventrolateral prefrontal cortex.

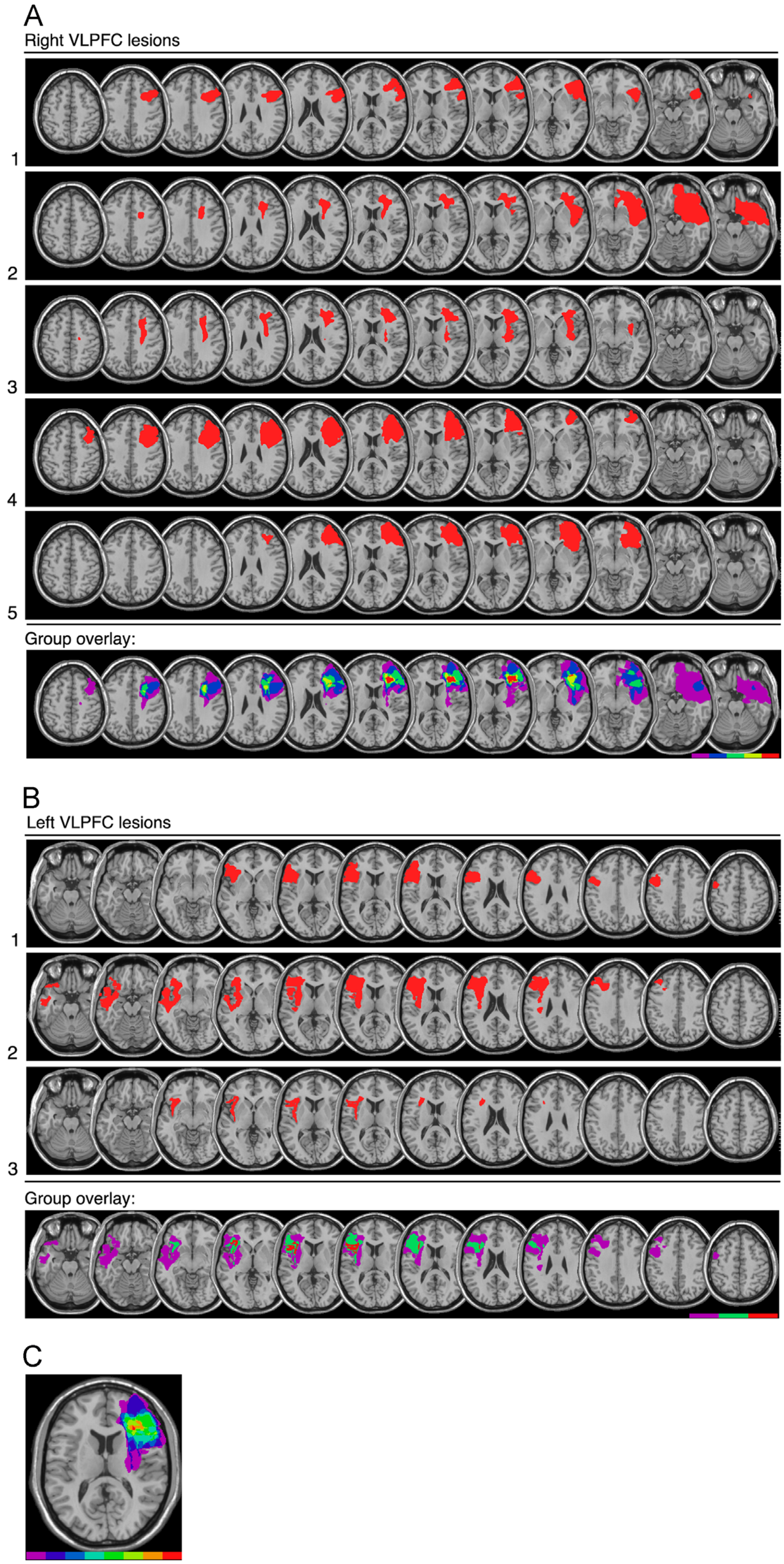

Fig. 1. Lesion reconstruction of the right (A) and left (B) VLPFC groups in the main study. Representative axial slices in standard space showing the location of damage in individual patients (row 1–5 in (A); row 1–3 in (B)) and extent of lesion overlap within each group (bottom row). In the group overlay, the region in red corresponds to the area of damage common amongst all subjects within a group. Left VLPFC lesions were reflected into the right hemisphere and added to a map of right VLPFC patients to show the area of lesion overlap, regardless of hemisphere, with red indicating the region affected in all patients, LVLPFC and RVLPFC, in the main experiment (C). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Patient demographic information and performance on aphasia and neglect screening tests (mean, range).

	RVLPFC (N=5)	LVLPFC (N=3)	Controls (N=21)
Age	55.8 (32–76)	51.7 (39–76)	58.9 (28–78)
Years of education	14.5 (7.5–18)	14.8 (12–17)	14.6 (11–20)
Beck depression inventory	11.6 (4–18)	18.7 (7–25)	
Sentence comprehension	0.95 (0.8–1)	0.87 (0.8–0.9)	
Naming	0.91 (0.8–1)	0.80 (0.7–0.9)	
Animal fluency[a]	17.2 (14–20)	13.7 (11–17)	
Letter fluency[a]	38.8 (11–53)	17.0 (6–26)	
Digit span forward	6.0 (5–7)	5.3 (4–7)	
Digit span backwards[b]	5.4 (4–7)	3.7 (3–4)	
Letter cancelation accuracy in left hemifield	0.97 (0.93–1)	0.99 (0.97–1)	
Letter cancelation accuracy in right hemifield	0.97 (0.93–1)	0.99 (0.97–1)	

[a] Between-group trends ($P<0.10$).
[b] Statistically significant between-group differences ($P<0.05$).

digit span (U=1.0; P=0.046) compared to RVLPFC. Controls scored above 25/30 on the Montreal Cognitive Assessment (Nasreddine, et al., 2005), did not take psychoactive medication and had no history of neurological or psychiatric illness. All participants passed a screening neurological exam including examination of visual fields, hearing and motor function. All participants also provided written, informed consent prior to participation in the study, in accordance with the Declaration of Helsinki. Participants received a nominal fee in compensation for their time. The Montreal Neurological Institute research ethics board approved the study protocol.

2.2. Participants in Experiment 2

As is often the case in human lesion studies of very specific anatomical regions-of-interest, Experiment 1 involved a relatively small sample. We therefore augmented these observations in an additional dataset of 12 right-handed patients with LVLPFC lesions and 27 age-matched healthy controls who had completed either a Stroop or an Eriksen flanker task gathered independently at another institution as part of unrelated work. Although there are some differences in the characterization and task formats, this dataset provided an important opportunity to test the replicability of the observed LVLPFC effects in Experiment 1. Eight LVLPFC patients (6 men and 2 women) and 15 controls (7 men and 8 women) completed a Stroop task. One patient was excluded for poor color naming ability, leaving 7 LVLPFC patients (5 men and 2 women). This patient exhibited a color anomia, such that colors were misnamed (or missed) at a high rate even under the neutral and letter string conditions (error rates of 28% and 20%, respectively). Inclusion of this subject did not alter the statistical significance of our findings. Nine LVLPFC patients (5 men and 4 women) and 12 controls (8 men and 4 women) completed the flanker task. Four of the patients were tested in both experiments (4 years apart) but no controls participated in both studies; therefore, the Stroop and flanker in Experiment 2 were analyzed separately. All lesions were caused by infarction in the precentral branch of the middle cerebral artery. Individual patient lesion characteristics are shown in Fig. 2. Patients with lacunar infarcts, white matter hyperintensities, significant medical complications, psychiatric disturbances, substance abuse, multiple neurological events or dementia were excluded. There was no significant difference between patient groups and controls with respect to mean age in the Stroop (P=0.67) or flanker study (P=0.75). Mean years of education also did not differ between patients and controls in the Stroop experiment (mean LVLPFC=16.7, controls=13.6; U=27.0, P=0.0639) or flanker study (mean LVLPFC=17.0, controls=16.6; U=46.5, P=0.5888). The trend for higher education in the Stroop patient group would bias towards a null result. Other demographic details and Western Aphasia Battery (WAB) scores for the patients are shown in Table 3. All participants provided written, informed consent prior to participation in the study, in accordance with the Declaration of Helsinki. Participants received a nominal fee in compensation for their time. The Institutional Review Board of the VA Northern California Health Care System approved the study protocol.

Table 3
Experiment 2: LVLPFC subject characteristics (mean ± SD).

	Stroop study (N=7)	Flanker study (N=9)
Age	56.7 ± 9.1	58.7 ± 11.5
Years of education	16.7 ± 3.6	17.0 ± 2.3
Lesion volume (cc)	82.0 ± 53.2	111.2 ± 67.4
Years since injury	5.6 ± 2.5	9.9 ± 3.4
Aphasia type (WAB)	WNL=3 Anomic=3 Unclassified=1	WNL=2 Anomic=3 Unclassified=3 Broca=1
WAB fluency	8.7 ± 1.7	7.4 ± 3.0
WAB comprehension	9.6 ± 0.5	9.5 ± 0.5
WAB repetition	9.1 ± 1.0	8.2 ± 1.8
WAB naming	8.7 ± 1.2	8.8 ± 0.9

WAB=Western Aphasia Battery; WNL=within normal limits.

2.3. Lesion analysis

Patient inclusion was based on damage to VLPFC, the anatomical boundaries of which encompass the inferior frontal gyrus ventral to the inferior frontal sulcus, anterior to premotor cortex (Brodmann's area [BA] 6), and posterior to the frontal pole (BA 10). VLPFC consists of three subdivisions (pars opercularis, pars triangularis, and pars orbitalis) corresponding approximately to BA 44, 45, and 47 respectively (Petrides and Pandya, 2002). A detailed description of image acquisition parameters and lesion reconstruction for Experiment 1 can be found in Supplementary materials.

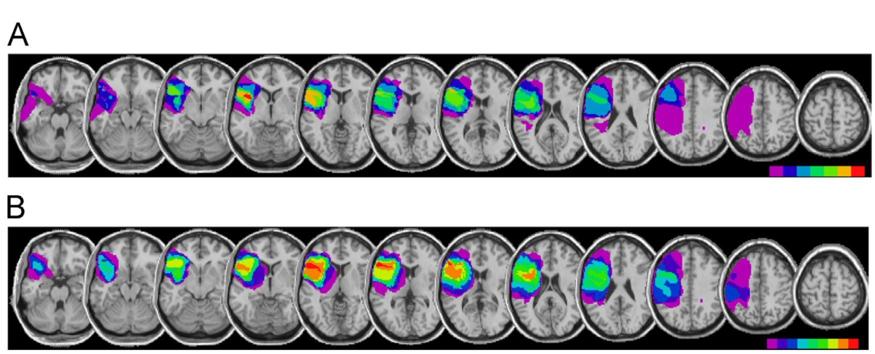

Fig. 2. LVLPFC group overlay for patients in the Stroop (A) and flanker (B) in Experiment 2. The region in red corresponds to damage common amongst all participants within each group. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.4. Tasks

Participants performed computerized versions of two classic tests of interference control: the color-word Stroop (Stroop, 1935) and Eriksen flanker tasks (Eriksen and Eriksen, 1974). These tasks require facilitation of goal-relevant features and inhibition of competing, goal-irrelevant dimensions. The primary measure of interest in both tasks was the interference effect, i.e. the reaction time (RT) difference between correctly performed incongruent and congruent trials. We did not include a neutral condition in either task in Experiment 1, and thus cannot comment on the relative contribution of facilitation and inhibition to the overall interference effect. A complete description of the Stroop and flanker tasks used in Experiment 2 can be found in Supplementary materials.

2.4.1. Stroop color naming task

The Stroop task used in Experiment 1, based on Fellows and Farah (2005), required participants to say aloud the ink color in which one of five color words was printed (red, blue, green, brown, or purple), inhibiting the prepotent tendency to read the words. All words were the names of the same five colors; thus, all trials were either congruent (e.g. the word "blue" printed in blue ink) or incongruent (e.g. the word "blue" printed in red ink). Stimuli were presented one at a time in the center of a black background in a pseudorandom order, and remained on the computer screen until the subject responded, followed by an inter-stimulus interval (ISI) of 1000 ms. A microphone connected to a response box (www.pstnet.com) recorded the onset of the verbal response. Forty practice trials, with an equal number of congruent and incongruent stimuli, were provided at the beginning of the task and there was a rest period after the first two blocks. The design of the Stroop task also allowed us to examine modulation of the interference effect by trial context, a measure that has been proposed as an additional index of cognitive control; participants completed four blocks of 50 trials, with two blocks of 20% incongruent trials and two blocks of 80% incongruent trials interleaved, a method employed in previous studies of conflict monitoring (Fellows and Farah, 2005; Swick and Jovanovic, 2002).

2.4.2. Eriksen flanker task

Participants performed a speeded version of the Eriksen flanker task on a laptop computer programmed in E-Prime 1 for the main experiment. Participants viewed a horizontal series of five arrows and were asked to indicate the direction of the central arrow as quickly and accurately as possible while ignoring flanking congruent ($<<<<<$ or $>>>>>$) or incongruent ($<<><<$ or $>><>>$) arrows. Speed and accuracy were equally emphasized. Participants used the index and middle finger of their dominant hand to press a right or left arrow key of a PST serial response box (http://www.pstnet.com). There were 400 trials, half congruent and half incongruent, with trial type varying according to a fixed, pseudorandom sequence. Ten practice trials were provided at the outset and participants received feedback about their reaction time after every 20 trials. The arrows remained on the screen for 100 ms and participants had 1000 ms to respond. A blank screen appeared after 1000 ms elapsed or immediately following response registration. The inter-stimulus interval was 200 ms.

2.4.3. Behavioral analyses

To prevent extreme RTs from unduly influencing the means for each participant, outliers (>3SD from the mean individual RT or under 100 ms) were excluded from the raw data resulting in a mean data reduction of 1.8% for flanker and 0.02% for the Stroop tasks. The nonparametric Wilcoxon signed-rank test was used for within-subject comparisons (i.e., to compare mean RT or error rate between congruent and incongruent conditions). To test whether RVLPFC or LVLPFC is critical in resolving conflict in the Stroop and flanker tasks, non-parametric, between-subject comparisons were performed on the interference effects in each task, here defined as the difference between mean incongruent and congruent RT. For three-group comparisons (Experiment 1), the Kruskal–Wallis test was used, followed by post-hoc Dunn's tests where appropriate (for similar analytic approach in other small sample human lesion studies, see Hilz et al., 2006; Petrides, 1985; Stepankova et al., 2004). The Mann–Whitney test was used for two-group comparisons (Experiment 2). To investigate whether a double dissociation exists between lateralized brain areas, we calculated z-scores of RT interference effects for both Stroop and flanker tests, based on the performance of controls and then tested for a significant interaction between lesion laterality and interference effects in the two tests, with the nonparametric repeated-measures Wald-type statistic, as implemented in the R-based statistical analysis package, *nparLD* (Noguchi et al., 2012). Unless otherwise stated, values are mean ± standard deviation. The level of statistical significance was set at $P<0.05$.

3. Results

3.1. Experiment 1

Responses on incongruent trials were slower compared to congruent trials among all participants for Stroop (Wilcoxon test, controls: $Z=-4.02$, $P<0.0001$; patients: $Z=-2.52$, $P=0.0117$) and flanker tasks (Wilcoxon test, controls: $Z=-4.02$, $P<0.0001$; patients: $Z=-2.52$, $P=0.0117$), resulting in identical statistical values in both tasks. Errors were also more frequent on incongruent compared to congruent trials in Stroop (Wilcoxon test, controls: $Z=-3.82$, $P=0.0001$; patients: $Z=-2.37$, $P=0.018$) and flanker tasks (Wilcoxon test, controls: $Z=-4.02$, $P<0.0001$; patients: $Z=-2.38$, $P=0.017$), but did not differ significantly between groups in either Stroop (Kruskal–Wallis, H (2)=0.32, $P=0.85$) or flanker tasks (Kruskal–Wallis, H (2)=0.6, $P=0.75$). Individual error rates and RTs are reported in Table 4.

In the Stroop task, critically, there was a significant difference in interference effect between groups (controls=169.7 ± 50.4, LVLPFC= 337.6 ± 97.9, RVLPFC=210.2 ± 86.4; Kruskal–Wallis, H (2)=6.836, $P=0.033$) (Fig. 3A). *Post-hoc* Dunn's tests showed that the LVLPFC group had a significantly larger Stroop effect than controls and the RVLPFC group (Fig. 3C, Table 5). Conversely, RVLPFC subjects did not differ from controls (Table 5). The exaggerated Stroop effect after LVLPFC damage was particularly striking when response conflict was greater (i.e. in blocks when incongruent trials were infrequent), with a trend towards an increased difference in Stroop interference effect between high conflict vs. low conflict blocks (Kruskal–Wallis, H (2)= 5.722, $P=0.057$; Dunn test LVLPFC vs. HC $P=0.002$, LVLPFC vs. RVLPFC $P=0.008$, RVLPFC vs. HC $P=0.90$).

The flanker interference effect also differed by group (controls=31.9 ± 11.3, LVLPFC=23.7 ± 16.9, RVLPFC=50.9 ± 7.8; Kruskal–Wallis, H (2)=9.593, $P=0.008$)(Fig. 3B): an identical Dunn

Table 4
Experiment 1: individual patient and group errors and reaction times (ms) in the Stroop and flanker tasks (Mean ± SD).

	(*Errors*)				(*Reaction times*)			
	Stroop I	*Stroop C*	*Flanker I*	*Flanker C*	*Stroop I*	*Stroop C*	*Flanker I*	*Flanker C*
RVLPFC 1	11	0	28	21	940.6	654.4	514.6	457.9
RVLPFC 2	0	0	8	10	762.8	624.7	507.9	463.3
RVLPFC 3	4	0	19	11	1167.0	845.9	574.0	524.2
RVLPFC 4	2	0	26	9	1003.8	852.1	517.4	475.0
RVLPFC 5	12	0	25	12	606.5	452.4	514.2	475.0
RVLPFC group	5.8 ± 5.4	0 ± 0	21.2 ± 8.1	12.6 ± 4.8	896.1 ± 217.1	685.9 ± 167.7	525.6 ± 27.3	474.8 ± 28.8
LVLPFC 1	4	0	19	6	1647.2	1208.0	545.2	537.7
LVLPFC 2	2	0	8	4	1232.5	988.6	551.8	529.4
LVLPFC 3	21	2	22	16	946.7	617.1	588.9	547.7
LVLPFC group	9.0 ± 10.4	0.7 ± 1.2	16.3 ± 7.4	8.7 ± 6.4	1275.4 ± 352.2	937.9 ± 298.7	562.0 ± 23.6	538.3 ± 9.2
Controls	4.5 ± 3.3	0.0 ± 0.2	26.5 ± 25.3	14.8 ± 22.0	840.7 ± 121.4	671.0 ± 107.7	469.2 ± 60.9	437.3 ± 59.2

I=Incongruent, C=Congruent, RVLPFC=Right ventrolateral prefrontal cortex, LVLPFC=Left ventrolateral prefrontal cortex.

Table 5
Experiment 1: *post-hoc* Dunn pairwise comparison of interference effect in ms (incongruent RT – congruent RT).

	Comparison	*Difference*	*SE Difference*	*P value*
Stroop effect				
LVLPFC vs. controls	167.85	38.22	0.0002[a]	LVLPFC > controls
LVLPFC vs. RVLPFC	127.33	45.22	0.0092[a]	LVLPFC > RVLPFC
RVLPFC vs. controls	40.53	30.81	0.1999	
Flanker effect				
RVLPFC vs. controls	18.96	5.67	0.0025[a]	RVLPFC > controls
RVLPFC vs. LVLPFC	27.18	8.32	0.0031[a]	RVLPFC > LVLPFC
LVLPFC vs. controls	8.22	7.03	0.2532	

[a] Significant difference ($\alpha=0.05$). SE=standard error; RVLPFC=right ventrolateral prefrontal cortex; LVLPFC=left ventrolateral prefrontal cortex; RT=reaction time.

post-hoc analysis showed a significantly larger flanker effect in RVLPFC patients compared to controls or LVLPFC subjects, with no difference in flanker effect between LVLPFC and controls (Fig. 3C, Table 5). LVLPFC participants were slower than controls across all trials (combined mean RT) in the flanker task (Kruskal–Wallis, H (2)=8.71, P=0.013; Dunn test LVLPFC vs. HC P=0.007) and showed a trend towards being slower on all trials in the Stroop task (Kruskal–Wallis, H (2)=4.63, P=0.099).

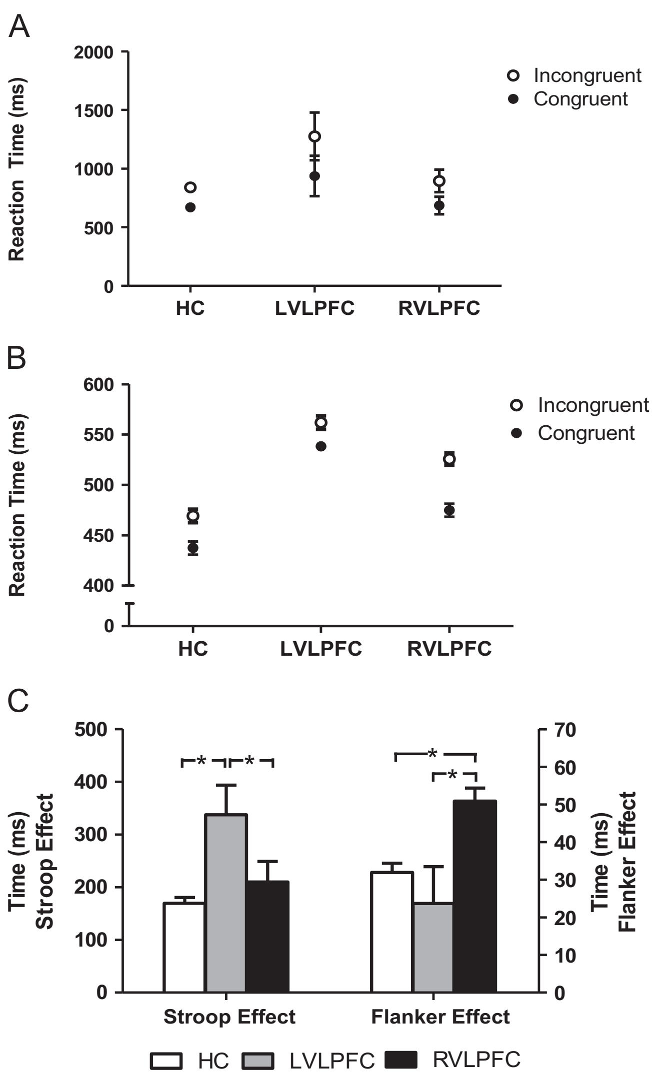

Fig. 3. Task performance expressed as mean RT for incongruent and congruent trials, by group, for the Stroop (A) and the flanker task (B) in the Experiment 1. Panel (C) shows the mean Stroop and flanker effect by group. Error bars indicate SEM. $^*P<0.05$, significant differences for pairwise comparisons (Dunn test).

To directly compare LVLPFC and RVLPFC interference control in both tasks and to test for a double dissociation, individual z-scores were calculated for the Stroop and flanker effect for each patient. The interaction between patient group and z-transformed task-specific interference effect was significant (Fig. 4), whether tested with a nonparametric repeated-measures *nparLD* (Wald-type statistic=10.2, P=0.001), or with parametric statistics (not shown).

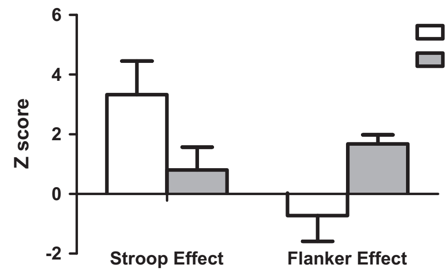

Fig. 4. Mean Stroop and flanker effects by group expressed as z-scores based on healthy control performance.

3.2. Experiment 2

As expected, responses on incongruent trials were slower compared to congruent trials among all participants in the Stroop (Wilcoxon test, controls: $Z=-3.41$, P=0.0007; patients: $Z=-2.37$, P=0.018) and flanker experiment (Wilcoxon test, controls: $Z=-3.06$, P=0.002; patients: $Z=-2.67$, P=0.008). Errors were also more frequent on incongruent trials in both tasks, but did not differ across groups in either the Stroop (controls=3.99 ± 4.1, LVLPFC=4.02 ± 4.22; Mann–Whitney, U=49.0, P=0.8) or flanker task (controls=3.99 ± 1.06, LVLPFC=3.81 ± 1.3; Mann–Whitney, U=48.0, P=0.67).

The LVLPFC group had a significantly greater Stroop interference effect compared to the control group (Mann–Whitney test, U=21.0, P=0.026). This effect could not be explained by aphasia: there was also a significant effect of group when examining only the three LVLPFC patients who performed within normal limits on the WAB (U=5.0, P=0.038) who, unlike the LVLPFC group as a whole, were not slower overall compared to controls (U=16.0, P=0.44). Conversely, the flanker experiment revealed no group difference for the interference effect (U=41.0, P=0.36). The mean RT for each condition is reported in Table 6.

Table 6
Experiment 2: reaction times (ms) in the Stroop and flanker tasks (mean ± SD).

Group	(*Reaction times*)			
	Stroop I	*Stroop C*	*Flanker I*	*Flanker C*
LVLPFC	1119.7 ± 279.9	865.3 ± 207.4	673.5 ± 127.2	608.2 ± 123.7
Controls	760.0 ± 122.1	636.2 ± 105.9	615.2 ± 143.8	559.2 ± 143.6

I=Incongruent, C=Congruent, LVLPFC=Left ventrolateral prefrontal cortex.

4. Discussion

We identified a double dissociation in contributions of right and left VLPFC to two classic tests of interference resolution in patients with damage centered on either region: intact right VLPFC was critical to resolve flanker, but not Stroop, interference while left VLPFC showed the opposite pattern. The single dissociation in Stroop and flanker performance after LVLPFC damage was replicated in two independent samples, gathered at a separate institution. This suggests that Stroop and flanker tasks rely differentially on prefrontal subregions, and left and right VLPFC make distinct contributions to interference control in humans.

Previous lesion studies examining the Stroop effect in larger groups of patients with damage to a range of PFC regions found mixed results (Perret, 1974; Stuss et al., 2001). Consistent with the results here, Perret (1974) showed that exaggerated Stroop interference was related to left PFC injury, but Stuss et al. (2001) found that this effect was seen only in patients with bilateral superior medial frontal damage. The latter study included only two patients with left VLPFC damage, out of a total sample of 51. Lovstad et al. (2012) found that half of a group of patients with mixed right and left lateral PFC injuries performed worse than controls and orbitofrontal patients on the color-word interference condition from the Delis–Kaplan Executive Function System test. However, the authors did not report the lesion location of participants with an exaggerated interference effect.

Unlike these PFC lesion studies, large-scale voxel-based lesion-symptom mapping (VLSM) studies do not make a priori assumptions about structure–function relationships. A study including 344 patients (165 involving PFC) found that ability to resolve Stroop interference localized to the left DLPFC (Gläscher et al., 2012). In addition, this study showed that another task loading on verbal control (controlled oral word association test) also depended on a left lateralized PFC locus. Previously, we performed a VLSM analysis in a group of 45 PFC patients, including some of the patients reported here, which demonstrated that left VLPFC damage was associated with exaggerated Stroop interference (Tsuchida and Fellows, 2013).

Unitary accounts of prefrontal organization suggest that PFC performs a set of general computations adapted to material according to context. A model proposed by Duncan and Miller (2002) suggests that PFC regulates behavior by flexibly coding any modality of task-relevant information, biasing competing representations to favor inputs relevant to current behavior. In a meta-analysis, Duncan and Owen (2000) postulate that co-recruitment of mid-dorsolateral, mid-ventrolateral and dorsal anterior cingulate regions does not exclude the possibility of finer specialization within this network. They suggest that specialization may be in "degree rather than kind", and relative specialization may occur, such as relatively stronger left PFC recruitment during verbal tasks. Relatedly, there is evidence that increasing task demands result in more distributed network representation (Woolgar et al., 2011). This claim would predict that greater conflict should increasingly recruit bilateral VLPFC. Contrary to this prediction, we observed that patients with left, but not right, VLPFC shows relatively increased interference during high conflict Stroop blocks, when incongruent stimuli were infrequent. Overall, our findings argue against domain-general models of prefrontal organization as they apply to the role of human VLPFC in interference control.

In contrast to the domain-general view, others suggest that PFC carries out a set of specific, highly localized functions. Recent models propose dorsal-ventral (Petrides, 2005), or rostral-caudal specialization (Badre and D'Esposito, 2009; Koechlin and Summerfield, 2007), suggesting that rostral PFC integrates increasingly abstract relationships between item categories whereas caudal PFC processes specific sensorimotor associations (Badre and D'Esposito, 2009; Badre et al., 2009). Notably, these models are material-general, emphasizing computations rather than content. Hemispheric distinctions in hierarchical PFC organization are largely unexplored (Levy and Wagner, 2011) and left/right VLPFC is often considered as a single functional unit (Koechlin and Jubault, 2006). Human fMRI and lesion studies propose an alternative interpretation of the rostro-caudal gradient with respect to domain generality: rostral PFC maintains information across multiple domains whereas content-based distinctions are represented in caudal PFC regions (Badre and D'Esposito, 2009; Rowe et al., 2007; Sakai and Passingham, 2003). The double dissociation we identified argues that laterality, in addition to position along the rostro-caudal axis, influences whether a PFC region is critically engaged in a given task. This is supported by human fMRI experiments showing material-specific preparatory activation in right or left VLPFC prior to a spatial or verbal working memory task respectively, whereas rostral PFC is activated under both conditions (Sakai and Passingham, 2003). Anatomical evidence is also consistent with this interpretation: monkey autoradiographic studies show that VLPFC, unlike rostral PFC (BA 10), receives projections from posterior primary sensory regions including parietal, occipital and inferotemporal cortex (Badre and D'Esposito, 2009; Petrides, 2005; Petrides and Pandya, 2009). In contrast, inputs to rostral PFC arise in multisensory areas, permitting control over integrated cognitive processes (Petrides and Pandya, 2007). Extending these findings, the double dissociation we observed in VLPFC (primarily BA 45, 47), argues that control signals in this region act upon task representations originating in the same hemisphere.

An older literature, stretching back decades, suggests that in humans left and right PFC have material-specific functions, with the former important for processing verbal, and the latter, visuospatial information (Jones-Gotman and Milner, 1977). There is converging support that this claim applies specifically to VLPFC: human lesion and neuroimaging studies show that right VLPFC is critical for spatial attention (Coulthard et al., 2008; Fink et al., 1997; Husain and Kennard, 1996; Urbanski et al., 2008; Vallar and Perani, 1987) while left VLPFC (notably Broca's area) is implicated in resolving interference between competing phonological, syntactic or semantic representations (Badre and Wagner, 2002; January et al., 2009; Kan and Thompson-Schill, 2004; Thompson-Schill et al., 1998). Moreover, an fMRI study by Stephan et al. (2003) showed that content and processing in higher-order cognitive tasks are localized to the same hemisphere, which, intriguingly, may develop through childhood (Bunge et al., 2002). Additional support for lateralized PFC contribution to cognitive control comes from fMRI on various forms of memory (Badre and Wagner, 2002; Feredoes et al., 2006; Golby et al., 2001; Jonides et al., 1993; Kelley et al., 1998; Wagner et al., 1998); studies have shown a region by condition interaction (Golby et al., 2001) where left VLPFC is activated in the presence of interfering verbal memoranda, highlighting PFC hemispheric specialization in another higher-order cognitive domain. An alternative cognitive models suggests that right lateral PFC is critical for monitoring, defined as the process of checking a task over time for 'quality control', whereas left lateral PFC implements task setting, the ability to maintain a task context, or stimulus–response relationship (Stuss and Alexander, 2007). This model has been used to explain the left lateralized involvement in Stroop conflict (Gläsche, et al., 2012) but does not readily account for our findings since Stroop and flanker tasks load on both of these processes.

Do the lateralized effects of VLPFC damage we observed relate to local processing in PFC or to lesion-induced disruption of processing in more distant areas? A rapidly growing literature argues that VLPFC is a hub in a dynamic, large-scale frontoparietal network that includes DLPFC, ACC and lateral parietal cortices (Cole et al., 2013; Cole and Schneider, 2007; Duncan, 2001 Fox et al., 2005; Seeley et al., 2007; Yeo et al., 2011). Our findings demonstrate that VLPFC is critical for interference resolution, but do not address how this region may interact with other brain areas (e.g. Banich, 2009; Cole and Schneider, 2007; Milham et al., 2003). The critical contributions of these other regions to cognitive control remain a matter of debate. Lesion evidence on the role of dorsal ACC on Stroop performance, and on performance of related control-demanding tasks, is decidedly mixed (e.g. Critchley et al., 2003; Fellows and Farah, 2005; Floden et al., 2011; Lovstad et al., 2012). Similarly, whether material-specific impairments occur after lateralized DLPFC damage remains an open question (Vanderhasselt et al., 2009).

Other evidence suggests that cortical networks are composed of dissociable components that perform specialized functions (Andrews-Hanna et al., 2010; Cole and Schneider, 2007; Stiers et al., 2010). Studies employing multivoxel pattern analysis in frontoparietal network nodes suggest that content, along with other task dimensions, is robustly represented in VLPFC (Stiers et al., 2010). Relatedly, a meta-analysis of 34 neuroimaging studies of cognitive control over conflicting verbal and visuospatial representations identified lateralized fronto-parietal networks, including VLPFC, engaged in interference resolution (Roberts and Hall, 2008). Building on the traditional concept of diaschisis (the remote impact of a focal lesion), Gratton et al. (2012) have shown that localized brain injury causes large-scale disruption of functional brain networks. These findings, together with the present study, show the value of converging methods in understanding local and network contributions to complex processes such as interference control.

Our findings have limitations that will be important to address in future research. The Stroop and flanker tasks we chose are widely used, (Fellows and Farah, 2005; Modirrousta and Fellows, 2008; Swick and Jovanovic, 2002), but each task calls on multiple cognitive processes, making it difficult to definitively identify the factor responsible for the observed dissociation. Details of task design may be particularly important in studying cognitive control (Floden et al., 2011). In addition to the verbal and non-verbal distinction, the tasks may differ in other regards. Prior behavioral and fMRI studies show positive covariance of working memory (WM) capacity with response speed (Cavanagh, 1972; Prabhakaran et al., 2011; Salthouse, 1996) and efficient resolution of interference (Kane et al., 2001; Kane and Engle, 2003). In Experiment 1, LVLPFC patients demonstrated generalized slowing across all trials in the flanker and Stroop tasks and significantly lower backwards digit span compared to RVLPFC patients. Overall slowed RTs in LVLPFC patients may represent a deficit in “energizing” attention, a finding that has been previously described in patients with PFC damage (Alexander et al., 2005; Stuss et al., 2005). However, our observation that LVLPFC patients showed generalized slowing yet dissociable interference effects supports the independence of processing speed and interference control. Similarly, the presence of this dissociation despite lower verbal WM performance in the LVLPFC group is at least preliminary evidence that general WM capacity can be unlinked from interference control. Lesion studies support distinct left and right lateral PFC contributions to verbal (Tsuchida and Fellows, 2009) and spatial WM respectively (Bor et al., 2006). We did not administer a spatial WM span in this study, and so cannot comment on whether there may be a relationship between material-specific WM capacity and cognitive control. Future work will be needed to definitively establish the cognitive processes underlying the structure–function relationships we identified here.

Only a few PFC lesion studies have demonstrated double dissociations, primarily between orbital and lateral PFC in non-human primates (Dias et al., 1996). Indeed, the paucity of such data has been used as an argument in support of equipotential theories of cognitive control. The carefully matched patients in this study, with damage centered on VLPFC in each hemisphere, provide a strong test of such theories. This hypothesis-driven, focused design resulted in a relatively small sample size, but the observation of a statistically robust double dissociation guards against many of the potential pitfalls of small sample studies in neuropsychology (Fellows, 2012). Performance of the Stroop and flanker task was spared in the right and left VLPFC patient groups respectively, thereby acting as a strong negative control. The replication of these findings in a second, independent LVLPFC sample supports the validity and reliability of our main result. These findings are evidence for hemispheric specialization of the VLPFC contribution to resistance to interference, a core aspect of cognitive control. While PFC may have the potential to serve general functions, our results argue that adult human VLPFC, at least, becomes specialized for interference control of specific material.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.neuropsychologia.2014.09.024.

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